STAGE 1 – Data Acquisition & Annotation

Block 1.1: Thermal Data Acquisition
- Ceiling-mounted thermopile sensor
- Resolution: 114×40
- Single-person indoor recording
- Personal AAG dataset collection

Block 1.2: Dataset Preparation
- Frame extraction from raw sequences
- Train / validation split

Block 1.3: Manual Annotation
- Bounding box (single person)
- Exactly 7 keypoints
- Action label (walk / sit / fall)

STAGE 2 – Unified Model Design

Block 2.1: Feature Extraction
- Backbone:
- MobileVIT-S
- CSPNeXt-PAFPN
- Multi-scale feature

Block 2.2: Detection Head
- Anchor-free detection
- Bounding box regression

Block 2.3: Pose Estimation Head
- 7-keypoint prediction
- Visibility estimation

STAGE 3 – Inference & Edge Deployment

Block 2.4: Temporal Fall Classiffificatio
- Keypoint sequence modeling
- GRU-based
- Binary fall probability

STAGE 3 – Inference & Edge Deployment

Block 3.2: Scientific Visualization Output